PYTHON DESIGN PATTERNS

Software Engineering | Sample PDF document for the Digital Library demo

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Creational patterns

Singleton - ensure a class has only one instance. In this project, DatabaseManager is a singleton that exposes database configuration.

Factory Method - define an interface for creating objects, but let subclasses decide which class to instantiate. DocumentFactory returns PDFDocument or TextDocument depending on the file extension.

Structural patterns

Facade - provide a unified interface to a set of subsystems. LibraryFacade orchestrates borrow, return and document workflows.

Proxy - provide a surrogate that controls access to another object. DocumentProxy validates the reader’s loan before opening the real file.

Behavioral patterns

Observer - define a one-to-many dependency so that when one object changes state, all dependents are notified. BorrowService notifies AdminNotification and ReaderNotification observers.

Strategy - define a family of algorithms and make them interchangeable. SearchStrategy supports title, author, ISBN and category searches.

This sample PDF is intentionally short and is used only for demo.